SOC Week 2 Report:- Alert Management and Incident Response

- **Theoretical Knowledge-**

1. Alert Priority Levels-

- **Introduction:-**

  Alert priority levels help SOC teams decide how serious a security issue is and how quickly it needs attention. By setting priorities, analysts can focus on the most dangerous threats first, instead of spending time on less important alerts.

- **Priority Levels:-**

➢ **Critical:** Very serious issues like ransomware or data breaches that need immediate action.

➢ **High:** Major threats such as unauthorized access that could cause damage if not handled quickly.

➢ **Medium:** Suspicious activities that should be investigated but are not immediately dangerous.

➢ **Low:** Minor issues like port scans that usually don't pose an immediate risk.

- **Assignment Criteria:-**

  Alerts are prioritized based on:-

  ➢ Asset importance (production vs test systems) .

  ➢ Exploit likelihood (known vulnerabilities) .

  ➢ Business impact (financial or operational loss) .

  Example: A CVSS score of **9.8 (Log4Shell)** is classified as **Critical**.

- **Scoring Systems:-**

  The **CVSS (Common Vulnerability Scoring System)** is widely used to measure risk based on:-

  ➢ Base metrics .

- Temporal metrics .
- Environmental metrics.

- **Objective:-**

  The main goal is to analyze security alerts properly and assign priorities using standard methods so that the most important threats are handled first.

2. Incident Classification-

- **Introduction:-**

  Incident classification is used to group different security events into categories, which makes it easier to understand the problem and respond quickly.

- **Incident Categories:-**
  - Malware.
  - Phishing .
  - DDoS attacks .
  - Insider threats .
  - Data exfiltration.

- **Taxonomy Frameworks:-**

  Standard frameworks used:-
  - **MITRE ATT&CK** (e.g., T1566 – Phishing) .
  - ENISA Incident Taxonomy .
  - VERIS framework.

- **Contextual Metadata:-**

  Incidents are enriched with:-
  - Source IP address.
  - Timestamp .
  - Affected systems .
  - Indicators of Compromise (IOCs).

- **Objective:-**

  To improve investigation efficiency through proper classification and labeling.

3. Basic Incident Response-

- **Introduction:-**

  Incident response is a structured approach to detecting, managing, and resolving security incidents.

- **Incident Response Lifecycle:-**
  - **Preparation** – Create policies and tools .
  - **Identification** – Detect and analyze alerts .
  - **Containment** – Isolate affected systems .
  - **Eradication** – Remove threats .
  - **Recovery** – Restore normal operations .
  - **Lessons Learned** – Improve future response**.**

- **Procedures:-**
  - System isolation .
  - Evidence collection (memory dump, logs) .
  - Hashing for integrity .
  - Communication and reporting .
  - Automation using SOAR tools.

- **Objective:-**

  To respond effectively and minimize the impact of cyber incidents.

• Practical Application-

4. Alert Management Practice:-

- **Overview:-**

  In this task, I worked on managing and prioritizing security alerts. The goal was to understand how alerts are classified, prioritized, and handled in a SOC environment.

- **Alert Classification:-**

  I created a table in Google Sheets to classify alerts based on type, priority, and MITRE techniques. This helped in organizing alerts in a structured way.

Alert ID	Type	Priority	MITRE Tactic
1	Phishing	High	T1566
2	Malware	Critical	T1059
3	DDoS	High	T1498
4	Port Scan	Low	T1046
5	Ransomware	Critical	T1486
6	Brute Force	Medium	T1110

Fig.1:- Alert Table

- **Alert Prioritization:-**

  Different alerts were assigned priorities using CVSS scores. For example, critical vulnerabilities like Log4Shell were given the highest priority, while minor activities like port scans were marked as low priority.

- **Incident Ticket:-**

  An incident ticket was created for a ransomware attack scenario. It included details such as title, description, IP address, priority, and assigned analyst.

Incident Ticket

Title: [Critical] Ransomware on Server-X

Description:

A suspicious file named crypto_locker.exe was detected on Server-X.

The system showed signs of ransomware activity.

The source IP address associated with this activity is 192.168.1.50.

Priority: Critical

Assigned to: SOC Analyst

Status: Open

Fig: 2- Incident Ticket

- **Escalation:-**

  An escalation email was drafted to report a critical incident to the Tier 2 team. This helped in understanding how serious incidents are communicated in real-world scenarios.

Subject Escalation:- Critical Ransomware AlertEscalation: Critical Ransomware Alert

---

Dear Tier 2 Team,

A critical ransomware alert has been detected on Server-X. Indicators include the presence of crypto_locker.exe and suspicious activity from IP address 192.168.1.50. Immediate containment actions have been initiated, including isolating the affected system to prevent further spread.

Further investigation is required to determine the scope of the attack and identify any lateral movement within the network.

Please prioritize this incident and provide advanced analysis support at the earliest.

Regards,
SOC Analyst

Fig: 3- Escalation

5. Response Documentation:-

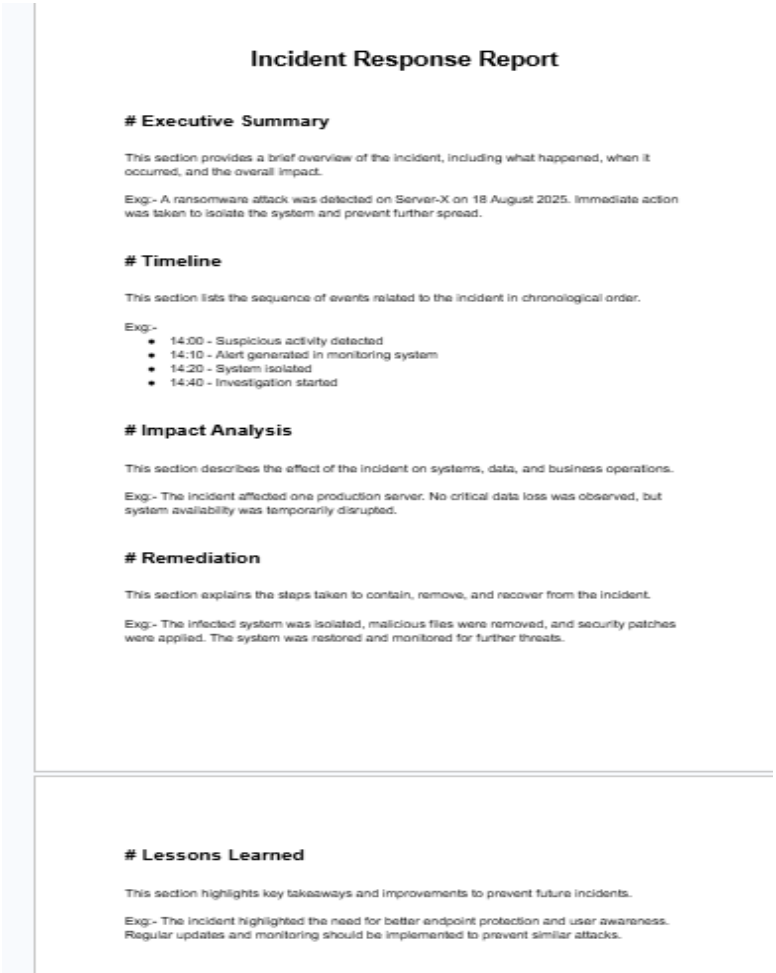
- **Overview:-**

  This section focused on documenting how incidents are handled step-by-step.

- **Incident Response Template:-**

  A structured report was created including:-

  - Executive Summary.
  - Timeline.
  - Impact Analysis.
  - Remediation.
  - Lessons Learned.



Incident Response Report

Executive Summary

This section provides a brief overview of the incident, including what happened, when it occurred, and the overall impact.

Exg:- A ransomware attack was detected on Server-X on 18 August 2025. Immediate action was taken to isolate the system and prevent further spread.

Timeline

This section lists the sequence of events related to the incident in chronological order.

Exg:-
- 14:00 - Suspicious activity detected
- 14:10 - Alert generated in monitoring system
- 14:20 - System isolated
- 14:40 - Investigation started

Impact Analysis

This section describes the effect of the incident on systems, data, and business operations.

Exg:- The incident affected one production server. No critical data loss was observed, but system availability was temporarily disrupted.

Remediation

This section explains the steps taken to contain, remove, and recover from the incident.

Exg:- The infected system was isolated, malicious files were removed, and security patches were applied. The system was restored and monitored for further threats.

Lessons Learned

This section highlights key takeaways and improvements to prevent future incidents.

Exg:- The incident highlighted the need for better endpoint protection and user awareness. Regular updates and monitoring should be implemented to prevent similar attacks.

Fig: 4- Incident Response Report

- **Investigation Steps:-**

A log was maintained to track actions taken during the incident, such as isolating the system and collecting data.

 # Lessons Learned

This section highlights key takeaways and improvements to prevent future incidents.

Exg:- The incident highlighted the need for better endpoint protection and user awareness. Regular updates and monitoring should be implemented to prevent similar attacks.

Investigation Log

Timestamp	Action
14:00	Isolated system
14:30	Collected logs
15:00	Memory dump collected
15:30	Malware analysis started

Fig: 5- Investigation Log

- **Phishing Checklist:-**

A checklist was created to verify phishing attacks, including checking email headers and suspicious links.

Phishing Checklist

- ☐ Check email header
- ☐ Verify link (VirusTotal)
- ☐ Identify affected users
- ☐ Block sender

This checklist is used to identify and respond to phishing attacks effectively.

Fig: 6- Phishing Checklist

- **Post-Mortem:-**

A short summary was written to identify what went wrong and how similar incidents can be prevented in the future.

Post-Mortem

Phishing attack occurred due to lack of user awareness. Improved email filtering and training are recommended. Early detection helped minimize damage. Future steps include implementing multi-factor authentication and stricter access controls.

Fig: 7- Post-Mortem

6. Alert Triage Practice:-

- **Overview:-**

  In this task, I analyzed alerts to determine whether they were real threats or false positives.

- **Triage Simulation:-**

  A sample alert (SSH brute-force attack) was analyzed with details such as source IP, priority, and status.

Alert Triage

Alert ID	Description	Source IP	Priority	Status
002	SSH Brute Force	192.168.1.100	Medium	Open

Threat Validation

Source IP 192.168.1.100 was checked in AlienVault OTX and found associated with brute-force attempts. Multiple reports confirm malicious behavior. This validates the alert as a true positive and requires blocking the IP and monitoring further attempts.

This process helps in identifying real threats and reducing false positives.

Fig: 8- Alert Triage

- **Threat Validation:-**

  The IP address was checked using threat intelligence sources, and it was confirmed to be malicious. This helped in understanding how real threats are validated.

7. Evidence Preservation:-

- **Overview:-**

  This task focused on collecting and preserving digital evidence during an incident.

- **Evidence Collection:-**

  A table was created to document evidence such as memory dumps, including details like date, collector, and hash value.

Evidence Preservation

Item	Description	Collected By	Date	Hash
Memory Dump	Server-X-dump	Analyst	2025-08-18	ABC123
Log Files	Server logs	Analyst	2025-08-18	XYZ789

This table represents the collected digital evidence during the incident. Hash values are used to ensure integrity and verify that the evidence has not been altered.

Fig: 9- Evidence Preservation

- **Importance of Hashing:-**

  Hash values were used to ensure that the evidence remains unchanged and maintains its integrity.

Alert	CVSS Score	Priority
Log4Shell	9.8	Critical
Port Scan	3	Low
Phishing	7.5	High
Malware	8.2	High
Brute Force	5.5	Medium
Info Leak	2.5	Low

Fig: 10- Priority Scoring

8. Capstone Project: Full Alert-to-Response Cycle:-

- **Overview:-**

  This was the final task where the complete incident lifecycle was simulated.

- **Attack Simulation:-**

  A cyber attack was simulated using a known vulnerability to understand how attacks occur.

- **Detection and Triage:-**

  The attack was detected, and an alert was generated with details such as timestamp, IP address, and MITRE technique.

- **Response:-**

  The system was isolated, the attacker's IP was blocked, and the system was monitored to ensure safety.

- **Reporting:-**

  A detailed report was prepared including summary, timeline, and recommendations.

- **Stakeholder Brief:-**

  A simple explanation of the incident was written for non-technical stakeholders.

Capstone Project: Full Alert-to-Response Cycle

Attack Simulation

A simulated cyber attack was performed using a known vulnerability in the vsftpd service on a Metasploitable2 system. The exploit was executed using Metasploit (exploit/unix/ftp/vsftpd_234_backdoor). This simulation was conducted to understand how real-world attacks are detected and handled in a SOC environment.

Detection and Triage

Timestamp	Source IP	Alert Description	MITRE Technique
2025-08-18 11:00:00	192.168.1.100	VSFTPD exploit	T1190

The alert was analyzed and classified as a high-priority incident based on its severity and potential impact.

Response

The affected virtual machine was immediately isolated from the network to prevent further spread of the attack. The attacker's IP address (192.168.1.100) was blocked using security controls. The system was then verified and monitored to ensure no further malicious activity was present.

Incident Report

Executive Summary:

A simulated cyber attack exploiting a vulnerability in the vsftpd service was conducted on a Metasploitable2 system. The attack was successfully detected and contained with no major impact.

Timeline:

At 11:00 AM, the attack was initiated from IP 192.168.1.100. The system detected unusual activity and generated an alert. Immediate action was taken to isolate the system.

Recommendations:

The incident highlights the importance of regular patching and monitoring. Implementing stronger access controls and continuous monitoring can help prevent similar attacks. Security awareness and timely response are critical to minimizing risks.

Stakeholder Brief

A security incident was detected involving an attempted attack on our system using a known vulnerability. The affected system was quickly isolated to prevent any spread. The attacker's IP address was successfully blocked, and no significant data loss occurred. Our team responded promptly and ensured that the system was restored safely. Additional security measures and monitoring have been implemented to prevent such incidents in the future.

Fig: 11- Capstone Project

- **Conclusion:-**

This report helped me understand how a Security Operations Center (SOC) works in handling security incidents. I learned how to classify and prioritize alerts, respond to incidents, and document each step properly.